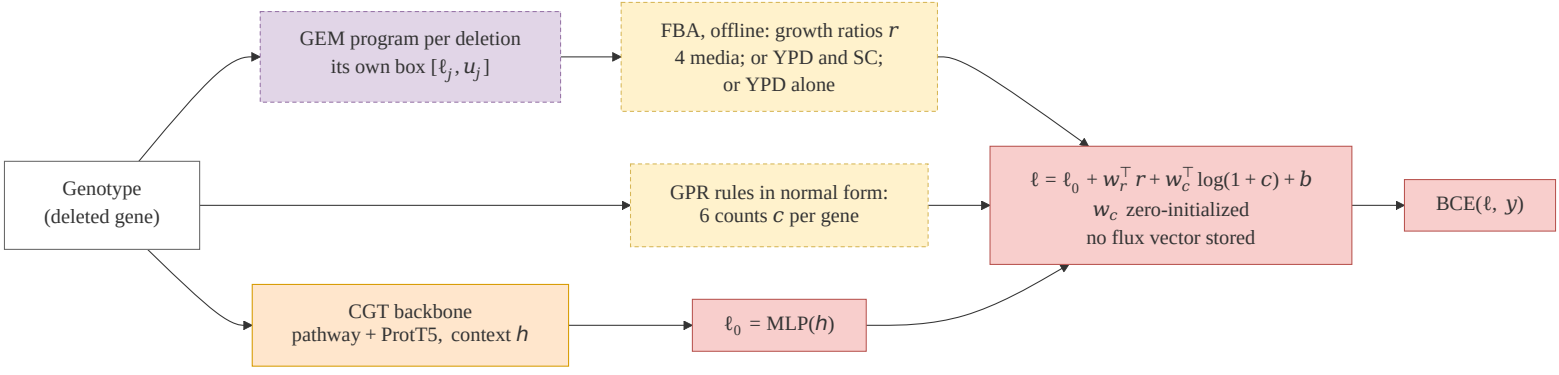

Genotype
(deleted gene)
GEM program per deletion
its own box $[\ell_j, u_j]$
FBA, offline: growth ratios r
4 media; or YPD and SC;
or YPD alone
GPR rules in normal form:
6 counts c per gene
CGT backbone
pathway + ProtT5, context h
ℓ_0 = MLP(h)
$\ell = \ell_0 + w_r^\top r + w_c^\top \log(1 + c) + b$
w_c zero-initialized
no flux vector stored
BCE(ℓ, y)